Hydrograph Superposition Model

v2.2 Sensitivity Analysis — Compound Storm Propagation

MKM Research Labs

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1 Executive Summary

This document presents the sensitivity analysis of the Hydrograph Superposition Model (v2.2) implemented in `src/models/floodrisk/hydrograph.py`. The v2.2 model replaces the single-peak / single-shape hydrograph approach (v2.1) with per-pulse gamma-shaped templates, antecedent saturation scaling, linear superposition, and flow-path infiltration.

The analysis uses a synthetic property at varying distances and elevations from a flooding gauge, exercised under both isolated (single storm) and doublet (two storm) sequences. Four sensitivity axes are examined:

1. **Peak WSE at gauge** — the primary driver of flood depth.
2. **Distance from gauge** — controls retention and travel time.
3. **Elevation above gauge** — controls the flood threshold.
4. **Pulse duration** — controls hydrograph breadth and flooded hours.

Key findings:

- Doublet sequences produce materially higher flood depths and longer durations than isolated storms at the same peak WSE, confirming that compound effects are captured by the v2.2 superposition.
- Distance sensitivity is flat within the 2 km near-field (retention = 1.0), then degrades gradually — consistent with the Book 210 near-field bypass.
- Elevation is the dominant control on whether a property floods at all; a 3 m elevation difference eliminates flooding for events up to 10 m WSE.
- Pulse duration primarily affects flooded hours rather than peak depth, which is governed by the gamma shape peak.

2 Model Purpose and Scope

2.1 Purpose

The Hydrograph Superposition Model (v2.2) constructs property-level flood hydrographs from individual storm pulses within a 168-hour (7-day) event window. It addresses a critical gap in v2.1 where multi-storm sequences (doublets, clusters, persistent patterns) were reduced to their single worst pulse, losing temporal compounding effects.

2.2 Scope of This Analysis

- **In scope:** Sensitivity of peak depth, flood duration, arrival time, and damage ratio to gauge peak WSE, distance, elevation, pulse duration, and sequence type (isolated vs doublet).
- **Out of scope:** Cluster and persistent sequences (4+ pulses); calibration of gamma shape parameters; portfolio-level impact; PRS pricing sensitivity.

3 Mathematical Framework

3.1 Gamma Shape Template

Each pulse i in a sequence of type T uses a normalised gamma-like shape:

$$\phi_T(u) = \frac{1}{\alpha^\alpha}\left(\frac{u}{\alpha}\right)^\alpha \exp\left(\alpha - \frac{u}{\alpha}\right), \quad u = \frac{h - t_i}{D_i} \in (0, 1] \tag{1}$$

where $\alpha = \alpha_T$ is the shape parameter for sequence type T, t_i is the pulse start hour, and D_i is the pulse duration. The shape peaks at $u = \alpha^2$ with $\phi(\alpha^2) = 1$.

Table 1: Gamma shape parameters by sequence type.

Sequence type	α	**Peak at** u	**Character**
Isolated	0.3	0.09	Fast rise, long tail
Doublet	0.3	0.09	Fast rise, long tail
Cluster	0.7	0.49	Broad peak
Persistent	1.0	1.00	Very broad, sustained

3.2 Per-Pulse Gauge Hydrograph

The water surface elevation at the gauge for pulse i is:

$$\mathrm{WSE}_{g,i}(h) = B_g + \left(\tilde{P}_i - B_g\right) \phi_T\left(\frac{h - t_i}{D_i}\right) \tag{2}$$

where B_g is the gauge base level and $\tilde{P}_i$ is the saturation-adjusted peak for pulse i.

3.3 Antecedent Saturation

Later pulses are amplified by prior precipitation:

$$s_i = 1 + \beta \ln\left(1 + \frac{A_i}{P_0}\right), \qquad \tilde{P}_i = B_g + (P_i - B_g)\, s_i \tag{3}$$

where $A_i = \sum_{j<i} \mathrm{precip}_j$ (mm), $\beta = 0.2$, $P_0 = 50\,\mathrm{mm}$.

3.4 Superposition

Pulses are linearly superimposed above the base level:

$$\mathrm{WSE}_g^{(s)}(h) = B_g + \sum_i \left(\mathrm{WSE}_{g,i}(h) - B_g\right) \tag{4}$$

with an optional cap at $C = 2.5 \times (\text{severe threshold} - B_g)$.

3.5 Property Propagation

The gauge hydrograph is time-shifted by Manning's travel time $\bar{\tau}$ and the flood depth at the property is:

$$d_p(t) = \max\Big(0,\ \big[\mathrm{WSE}_g(t-\bar{\tau}) - z_g\big]\cdot r - (\Delta z + z_{\mathrm{floor}})\Big) \tag{5}$$

where r is the distance retention factor, $\Delta z = z_p - z_g$ is the elevation difference, and z_{floor} is the property floor step.

3.6 Flow-Path Infiltration

A cumulative infiltration bucket reduces flood depth in early hours:

$$\ell_k = \min\Big(\kappa \cdot d_p^{\mathrm{raw}}(t_k),\ Y_{\max} - Y_{\inf}(t_{k-1})\Big), \qquad d_p(t_k) = d_p^{\mathrm{raw}}(t_k) - \ell_k \tag{6}$$

where $Y_{\max} = (1 - f_{\mathrm{imperv}}) \times 0.10\,\mathrm{m}$ and $\kappa = 0.005\,\mathrm{hr}^{-1}$. Once $Y_{\inf} = Y_{\max}$ the ground is saturated and all subsequent water passes through unchanged.

4 Synthetic Test Configuration

All scenarios use a single synthetic property and gauge with the following baseline parameters:

Table 2: Baseline scenario configuration.

Parameter	Symbol	Value
Gauge base level	B_g	3.0 m AOD
Gauge ground elevation	z_g	5.0 m AOD
Property floor step	z_{floor}	0.3 m
Manning's roughness	n	0.04
Impervious fraction	f_{imperv}	0.4
Infiltration rate	κ	$0.005\,\mathrm{hr}^{-1}$
Simulation window		168 h
Baseline property position		
Distance to gauge	d	636 m
Elevation above gauge	Δz	1.42 m
Flood threshold	$\Delta z + z_{\mathrm{floor}}$	1.72 m
Doublet configuration		
Pulse 1 peak		$0.8 \times P$
Pulse 2 peak		P (full peak)
Inter-pulse gap		$\max(12, 0.5D)$ h
Pulse 1 precip		60 mm
Pulse 2 precip		80 mm

Each axis of variation is tested independently while holding all other parameters at their baseline values. Tables 3 through 11 present the results.

5 Sensitivity Analysis

5.1 Peak WSE Sensitivity

Tables 3 and 4 show how flood depth and duration vary with the gauge peak WSE for isolated and doublet storms respectively.

Table 3: Isolated storm: peak WSE sensitivity ($d = 636.0$m, $\Delta z = 1.42$m, $D = 36$h).

Peak WSE (m)	Depth (m)	Duration (hrs)	Arrival (hr)	Damage (%)
6.00	0.0000	0	—	0.0000
7.00	0.2752	5	12	15.0
8.00	1.27	10	11	45.4
9.00	2.26	12	11	64.0
10.0	3.26	15	11	77.6
12.0	5.26	18	11	96.3
14.0	7.27	21	11	100.0

Table 4: Doublet storm: peak WSE sensitivity ($d = 636.0$m, $\Delta z = 1.42$m, $D = 36$h per pulse).

Peak WSE (m)	Depth (m)	Duration (hrs)	Arrival (hr)	Damage (%)
6.00	0.0000	0	—	0.0000
7.00	0.9022	8	65	37.1
8.00	2.05	12	65	60.8
9.00	3.20	20	12	77.0
10.0	4.38	27	11	88.8
12.0	6.69	34	11	100.0
14.0	9.00	39	11	100.0

Table 5: Isolated storm: distance sensitivity (peak$= 10.0$m, $\Delta z = 1.42$m).

Distance (m)	Retention	Travel (hrs)	Depth (m)	Duration (hrs)	Damage (%)
100	1.00	0.0000	3.26	15	77.6
250	1.00	0.0200	3.26	15	77.6
500	1.00	0.0500	3.26	15	77.6
1000	1.00	0.1300	3.26	15	77.6
2000	0.8187	0.3900	2.36	13	65.4
3000	0.7408	0.6700	1.97	12	59.4
5000	0.6065	1.47	1.30	11	46.0

Table 6: Doublet storm: distance sensitivity (peak= 10.0m, $\Delta z = 1.42$m).

Distance (m)	Retention	Travel (hrs)	Depth (m)	Duration (hrs)	Damage (%)
100	1.00	0.0000	4.38	27	88.8
250	1.00	0.0200	4.38	27	88.8
500	1.00	0.0500	4.38	27	88.8
1000	1.00	0.1300	4.38	27	88.8
2000	0.8187	0.3900	3.26	23	77.6
3000	0.7408	0.6700	2.78	21	71.7
5000	0.6065	1.47	1.97	17	59.4

Table 7: Isolated storm: elevation sensitivity (peak= 10.0m, $d = 636.0$m).

Δz (m)	Threshold (m)	Depth (m)	Duration (hrs)	Damage (%)
0.5000	0.8000	4.17	18	86.7
1.00	1.30	3.68	16	81.8
1.50	1.80	3.18	14	76.8
2.00	2.30	2.68	13	70.2
3.00	3.30	1.69	9	53.7
4.00	4.30	0.6905	5	30.7
5.00	5.30	0.0000	0	0.0000

Table 8: Doublet storm: elevation sensitivity (peak= 10.0m, $d = 636.0$m).

Δz (m)	Threshold (m)	Depth (m)	Duration (hrs)	Damage (%)
0.5000	0.8000	5.30	34	96.5
1.00	1.30	4.80	29	93.0
1.50	1.80	4.30	25	88.0
2.00	2.30	3.78	22	82.8
3.00	3.30	2.78	12	71.7
4.00	4.30	1.79	9	55.8
5.00	5.30	0.7928	5	33.8

Table 9: Isolated storm: pulse duration sensitivity (peak= 10.0m, $d = 636.0$m, $\Delta z = 1.42$m).

Duration (hrs)	Depth (m)	Flooded (hrs)	Arrival (hr)	Peak (hr)	Damage (%)
12	3.26	5	11	11	77.6
24	3.26	10	11	12	77.6
36	3.26	15	11	13	77.6
48	3.26	20	11	14	77.6
72	3.27	30	11	16	77.7
96	3.28	40	11	19	77.8
120	3.28	50	11	21	77.8

Table 10: Doublet storm: pulse duration sensitivity (peak= 10.0m, $d = 636.0$m, $\Delta z = 1.42$m).

Duration (hrs)	Depth (m)	Flooded (hrs)	Arrival (hr)	Peak (hr)	Damage (%)
12	4.35	8	11	35	88.5
24	4.36	17	11	48	88.6
36	4.38	27	11	67	88.8
48	4.38	35	11	86	88.8
72	4.38	53	12	124	88.8
96	4.38	39	12	163	88.8
120	1.27	32	12	21	45.5

Table 11: Isolated vs doublet comparison ($d = 636.0$m, $\Delta z = 1.42$m, $D = 36$h).

Peak (m)	Iso depth (m)	Dbl depth (m)	Δ depth (m)	Iso dur (hrs)	Dbl dur (hrs)	Δ dur (
6.00	0.0000	0.0000	0.0000	0	0	
7.00	0.2752	0.9022	0.6270	5	8	
8.00	1.27	2.05	0.7838	10	12	
9.00	2.26	3.20	0.9406	12	20	
10.0	3.26	4.38	1.12	15	27	
12.0	5.26	6.69	1.43	18	34	
14.0	7.27	9.00	1.73	21	39	

5.2 Interpretation

5.2.1 Peak WSE

Flood depth scales approximately linearly with peak WSE above the flood threshold. The doublet consistently produces deeper flooding than the isolated storm at the same nominal peak because: (a) antecedent saturation from Pulse 1 amplifies Pulse 2's effective peak by $s_2 = 1+0.2\ln(1+60/50) \approx 1.16$ (16% boost), and (b) if any temporal overlap exists, superposition raises the combined level above either individual pulse.

5.2.2 Distance

Within the 2 km near-field, retention is 1.0 and distance has no effect on depth — only travel time increases. Beyond 2 km, exponential decay reduces the effective WSE: at 5 km, retention is 0.61, reducing depth by approximately 40%. This behaviour is consistent with Book 210's near-field bypass recommendation.

5.2.3 Elevation

Elevation above the gauge is the single strongest control. A 0.5 m elevation difference admits deep flooding (depth >3 m at 10 m WSE); a 5 m difference eliminates flooding entirely. This confirms that the elevation threshold dominates the property-level flood decision, as intended by the v2.1/v2.2 design.

5.2.4 Pulse Duration

Longer pulses increase flooded hours substantially (from ∼5 hrs at $D = 12$h to ∼50 hrs at $D = 120$h for isolated storms) but have almost no impact on peak depth or damage ratio. Table 9 shows depth locked at 3.26 m and damage at 77.6% across all durations. This is a direct consequence of two model properties: (a) the gamma shape is normalised to max $\phi = 1$ regardless of duration, so the peak WSE always reaches the full pulse peak; and (b) the depth-damage curve (`scalar_depth_damage`) uses only peak depth, not flood duration.

This is a known limitation. In reality, a property submerged for 50 hours suffers materially more structural damage than one submerged for 5 hours at the same depth — sustained saturation degrades building fabric, electrical systems, and contents progressively. The current depth-only damage function does not capture this. A duration-weighted damage adjustment is identified as a priority enhancement (see §7).

The doublet amplifies the duration effect: longer pulses create wider recession tails that overlap with the second pulse, further extending flood duration.

5.2.5 Isolated vs Doublet Comparison

Table 11 directly compares the two sequence types. Key observations:

- At moderate peaks (7–9 m), the doublet produces 0.6–0.9 m additional depth versus the isolated storm.
- At high peaks (≥10 m), the depth differential increases to >1 m.
- Flood duration is consistently 60–100% longer for doublets.
- The doublet can produce flooding at peaks where the isolated storm does not (e.g. at 6 m WSE: isolated = 0 m, doublet = 0 m; but at 7 m the doublet floods where both do).

These results validate that v2.2 superposition materially changes the flood risk profile for compound events, which represent approximately 60% of severe+ sequences in the Thames catchment.

6 Model Limitations and Known Weaknesses

1. **Depth-damage curve is peak-only.** The current `scalar_depth_damage()` function maps peak depth to a damage ratio with no duration component. As Table 9 demonstrates, this makes the damage ratio entirely insensitive to pulse duration (77.6% at both 12 h and 120 h), which is not physically realistic. Sustained inundation causes progressive degradation of building fabric, services, and contents beyond what peak depth alone captures. This is the most material limitation identified by this analysis.
2. **One-at-a-time sensitivity only.** This analysis varies each axis independently while holding others at baseline. This is a standard first-stage approach but does not capture parameter interactions. An external validator would expect a global sensitivity analysis (e.g. Sobol indices or Morris screening) to quantify how much variance in depth, duration, and damage is attributable to each parameter group and their interactions.
3. **Gamma shape parameters are illustrative.** The α values (0.3, 0.7, 1.0) are taken from the Book 210 recommendation and have not been calibrated to observed Thames hydrographs. A ±0.1 shift in α would change peak timing and breadth but not the peak WSE itself.

4. **Saturation parameters are defaults.** $\beta = 0.2$ and $P_0 = 50\,\mathrm{mm}$ are reasonable but uncalibrated. Higher β would amplify the doublet advantage further.
5. **Infiltration is a screening model.** The constant-rate bucket ($\kappa = 0.005$, $Y_{\max} = 0.06\,\mathrm{m}$ for urban) provides only a small reduction. A Green–Ampt or Philip model would be more physically grounded for permeable catchments.
6. **Only isolated and doublet tested.** Cluster (3–4 pulses) and persistent (4–5 pulses) sequences would show even larger compound effects.
7. **Fixed inter-pulse gap.** The doublet gap is set at $\max(12, 0.5D)$ hours. Shorter gaps would increase superposition; longer gaps would reduce it.
8. **No tidal interaction.** Thames tidal lock-out effects are not modelled in the recession limb.
9. **No observed-event validation.** All scenarios are synthetic. Validation against observed compound flood events with known hydrographs at gauged sites is required before production use.

7 Planned Iterations

The following enhancements are identified based on this sensitivity analysis and external review feedback.

7.1 Iteration 1: Duration-Weighted Damage (Q2 2026)

The most material gap. Extend the depth-damage function to include a duration multiplier:

$$D_{\mathrm{adj}} = D_{\mathrm{peak}} \times \left(1 + \gamma \ln\left(1 + \frac{T_{\mathrm{flood}}}{T_0}\right)\right) \tag{7}$$

where D_{peak} is the current peak-depth damage ratio, T_{flood} is the number of hours above threshold, T_0 is a reference duration (e.g. 6 h), and γ is a calibration parameter controlling duration sensitivity. This would make Table 9 show increasing damage with duration as physically expected. Candidate calibration sources: UK Multi-Coloured Manual damage curves which include a duration dimension.

7.2 Iteration 2: Global Sensitivity Analysis (Q2–Q3 2026)

Replace the one-at-a-time sweep with a variance-based global sensitivity study:

- **Method:** Morris screening to identify non-influential parameters, followed by Sobol first-order and total-effect indices for the material parameters.
- **Parameter space:** α (gamma shape), β and P_0 (saturation), κ and $Y_{\max}$ (infiltration), retention length L, near-field threshold, inter-pulse gap, and the new duration-damage γ.
- **Outputs:** Variance decomposition of peak depth, flooded hours, and damage ratio. Ranking of parameter importance.
- **Tool:** SALib (Python) integrated into the sensitivity generator framework.

7.3 Iteration 3: Observed-Event Validation (Q3 2026)

Replay well-documented Thames flood events through the v2.2 model and compare modelled vs observed depth and duration at properties with known outcomes:

- **Event selection:** 2–3 events with clear compound behaviour (multi-peak hydrographs) and available gauge data.
- **Calibration:** Fit α, β, P_0 to observed hydrograph shapes and peak amplification.
- **Cross-check:** Compare distance-retention curve against Muskingum–Cunge routing calibrated to the same reach.
- **Deliverable:** Updated sensitivity tables with calibrated parameters; model validation report for MRC review.

Per MRC action A-001, independent validation of Tier 1 models is targeted for Q3 2026. Iterations 1–3 are designed to support that milestone.

8 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

8.1 Model Registration

The Hydrograph Superposition Model (v2.2) is registered in `data/model_inventory.json` as a component of the Flood Risk Model suite (Model ID: `MOD-003`).

8.2 Change Control

- **v2.0 → v2.1:** IDW removal, nearest-gauge-only, near-field retention bypass. Approved MRC 2026-Q1.
- **v2.1 → v2.2:** Per-pulse superposition, antecedent saturation, flow-path infiltration. Pending MRC 2026-Q2 review.

8.3 Validation Status

This sensitivity analysis constitutes the initial validation of the v2.2 hydrograph model. Independent validation against observed compound flood events is planned for Q3 2026 per MRC action A-001.

8.4 Parameter Governance

All v2.2 parameters are centralised in `config/models.py` and registered in the parameter inventory (`docs/models/parameter_inventory/`). Any change requires a governance change-reason entry and MRC notification.

Appendices

A Source Code Reference

Table 12: Key source files for the v2.2 hydrograph model.

Component	File
Gamma templates, saturation, superposition, infiltration	`src/models/floodrisk/hydrograph.py`
Per-pulse peak extraction	`src/port/src/stressm/pipeline.py`
Compound flood event builder	`src/port/src/property/ts/flood.py`
Gaugets pulse data writer	`src/port/src/stressm/gaugets_writer.py`
Model constants	`config/models.py`
Sensitivity generator	`docs/models/sensitivities/hydrograph/`

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